End of Year Competition

Mission Guide

2026

Integrated Descent Telemetry Satellite

STEC Initiative TMU CanSat

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Updates

Date	Update
March 12, 2026	Updated height satellite is raised to, PDR submission deadline changed, M8 requirement added, appendix added, clarifications added to budget.

Introduction

This Year's Mission

Design a satellite that consists of a payload connected to a drone which will transport the satellite up to a height of 120m. When the payload is raised to 120 m, a release mechanism will separate the payload from the drone after 5s.

The satellite shall descend at a speed of 3 m/s and shall determine the start of descent by detecting a decrease in altitude of at least 1 metre (1m) over three consecutive sensor readings using a barometric pressure sensor. Upon detecting descent, the CanSat shall transmit a single telemetry packet containing at minimum the following data: Altitude (m), GPS Position, Rotation (rev/s), Packet sequence number. The packet shall be sent at least 3 times.

During descent, the payload must maintain stability by having limited pitch and roll oscillations (no more than 45 degrees from the nadir) and minimal yaw rotation (no greater than 1 rev/s). Upon landing, the satellite must remain intact.

Competition Operation

The competition consists of two stages. Teams of 4-5 students will first submit a Preliminary Design Review (PDR) for evaluation. Secondary leads and team captains will review all PDRs, and the top 10 teams will advance to the second stage: the Manufacturing stage. In this stage, teams will build and test their satellites, ensuring they meet all requirements to maximize their final score.

Stage 0 – Team Registration

In order to participate in the End of Year Competition, those who wish to participate must form a group of 4-5 students. At a later date, the below form will be released. The deadline to register for the competition is **February 9, 2026.**

Stage 1 – PDR

The Preliminary Design Review (PDR) is a formal submission in which teams propose their satellite design to the judges, demonstrating that it is suitable for manufacturing and launch. Examples of past PDR presentations can be found on the CanSat website here: https://cansatcompetition.com/documents.html.

The PDR slide presentation requirements and the evaluation rubric are provided in the PDR section of this document.

Stage 2 – Manufacture and Launch

The scoresheet is included inside of the requirements for this competition. Groups work hard in order to satisfy as much of these requirements as possible during the actual deployment of the satellite. Check the requirements section of this document to review the scoresheet for this competition.

Timeline

Milestones and Dates

These milestones are subject to change, depending if an extension is needed for these key events.

Date	Milestone
February 9, 2026	The Google Form to sign up individually or as groups closes.
March 22 (11:59:59 PM)	Submission deadline for the PDR deadline.
March 30, 2025	Top 10 Teams list will be released. Number of teams subject to possible change/increase.

March 30 - May 29, 2026	Manufacturing Phase.
May 30, 2026	Launch of the payloads (competition day).

Requirements

The requirements of this competition are broken down into the following sections. These will determine the final score that each team will receive.

Operational Requirements

Requirement Number	Requirement
O1	The Satellite payload shall weigh no more than 500g.
O2	The Satellite payload shall be deployed via release mechanism.
O3	The Satellite shall be released at a height of 120m after 10s.
O4	The Satellite payload shall land upright or with high stability.
O5	The Satellite payload shall land intact (does not shatter or break).

Mechanical Requirements

Requirement Number	Requirement
M1	The Satellite shall descend at a speed of 3 m/s. Error is +/- 1 m/s.
M2	The Satellite shall have a width no larger than 125 mm ± 5%.
M3	The Satellite shall have a height no larger than 200 mm ± 5%.
M4	Payload must maintain stability during descent, maintaining limited pitch and roll oscillations of no more than 45 degrees from the nadir.

M5	Payload must maintain stability during descent, maintaining limited yaw rotation (no greater than 1 rev/s).
M6	Electrical and mechanical components shall be hard mounted safely and securely. ● Standoffs, screws, high performance adhesives etc. are permitted. ● Tape, rubber bands, etc. are not permitted.
M7	The satellite shall release from a U-bracket mount available on the drone.
M8	A parachute must be securely attached to the satellite at all times.

Electrical Requirements

Requirement Number	Requirement
E1	The Satellite shall take a minimum of 3 consecutive sensor readings.
E2	The Satellite shall determine the start of descent using a barometric pressure sensor.
E3	The Satellite shall measure its altitude using air pressure.
E4	The Satellite shall track its position using GPS, measure its acceleration and rotation rates.
E5	The power switch of the Satellite shall be easily accessible.
E6	Battery source may be alkaline, Ni-Cad, Ni-MH or Lithium. Lithium polymer batteries are not allowed. Coin cells are allowed.

Software Requirements

Requirement Number	Requirement
S1	Descent shall be detected when the altitude decreases by greater than or equal to 1m by using the sensor readings.

S2	Upon detecting descent, the Satellite shall transmit at least 1 telemetry packet, including information on the following: ● Altitude (m) ● GPS Position (latitude, longitude) ● Rotation rate (rev/s). ● Packet sequence number
S3	The Satellite shall transmit the telemetry packet at least three times to ensure successful reception.
S4	The ground station shall generate csv files of all sensor data as specified in the Telemetry Requirements section.
S5	The collected telemetry data (specified in S2) shall be plotted. All telemetry shall be displayed in the International System of Units (SI).
S6	All data shall be shown/displayed in ground station GUI.

Important Documents

Below are important resources and documents that will be needed to participate in the competition:

- **PDR Template** – Use this template to complete the requirement for Stage 1 **(March 2026)**.
- **Scoresheet** – Rubric for the competition. Determines the points assigned to each team **(February 2026)**.

Budget per Team

The budget, tools, and resources will be allocated to each team who is able to progress to Stage 2 of the Competition. This budget will be updated on further revisions and announced in the official Discord Server of the competition.

Below are the official line items that will be provided to each team. However, if a team identifies a comparable component that matches or is cheaper than the listed price of the item listed below, the leads may decide to replace that item for your team. This can include manufacturing methods of higher complexity that require different components to the ones listed below, if they are comparable or below the budget listed.

Additionally, if a physical component is self-bought or pre-owned, it may be used within certain restrictions. Pre-made flight controllers may not be purchased/used, while custom made PCB created by the team is allowed. The PCB must display the team name using a silkscreen layer. The cost of the pre-owned physical component may be used to transfer to another item with a limit of $100 of value. Additional items borrowed from TMU CanSat or other University resource (if permission to borrow is granted) can count towards this as well.

You may additionally request to book a budgeting meeting with the STECI leads after passing to Stage 2 of the Competition if your team has an interest in making any replacements. Tools for manufacturing (soldering station, cutting tools, etc) and fasteners or other materials (screws, bolts, wires, rope and nylon cord, fabric, etc) will be provided outside of the below budget. This budget may be subject to change with further revisions.

Equipment	Price per Team	Use
Arduino Uno Rev3 and Breadboard (x2 included)	~$40	Flight Computer
BMP280	~$7.5	Temperature and Atmospheric Pressure Sensor
NEO-6M	~$16	GPS
MPU-6050	~$4	Accelerometer and Gyroscope
NRF24L01 Set	~17	Antenna and Communications
3D Printing Filament	~$25-50	1-2 kg of Filament per team.
Micro Servos (SG90)	$3-6	Servos for actuation
Battery	$20	Power of the Satellite

Appendix A: Mounting

Your satellite design will be lifted into the air using a drone. Your satellite must be able to be **securely attached to the drone's u-bolt**. This u-bolt is provided by the STECI team, and this is the standard mounting that is applied to all teams to keep consistency. It acts as an accessible point for which your satellite can "hook" onto the drone. As per the mission statement, you must have your release mechanism activated to separate your satellite from the drone. The base of the drone, towards its center, will have the u-bolt **permanently attached.** The u-bolt cannot be tampered with or removed. The u-bolt will have an ID of 27mm, height of 40mm when installed onto the drone. Below is an image of a standard u-bolt:

You may set up or connect your release mechanism to the u-bolt before the liftoff however you prefer, as long as the connection is strong enough to ensure that it does not fall from the drone while it is being lifted (for safety).

Appendix B: Manufacturing

In terms of manufacturing, you will be expected to manufacture or partially manufacture the following parts of your satellite:

- **Parachute:** You must design and make your own parachute. The materials used to manufacture the parachute will be provided (such as fabric, nylon cord, etc).
- **Flight controller:** You are responsible for using the provided electronic components (or alternatives, as per the Budget section of this guide). The software can be developed by your team or be premade (for example, the use of open-source software).
- **Structure:** To manufacture the satellite, you may use 3D-printing with the provided material. Alternative materials or methods are allowed, but they will not be covered within your team's budget.
- **Descent mechanism:** You may choose your desired descent mechanism as long as you are additionally using a parachute for additional safety. The descent mechanism does not have to include reaching a target location.

Appendix C: Dimensional Requirements

Dimensional requirements apply to the main **payload system** of your satellite. The parachute is not included within the dimensional constraints (such as height and weight) of your satellite as long as the drone can reliably lift the satellite to the required height in the air. The payload system of your satellite is defined as the main structure that houses your electronics and release mechanism.

The release mechanism can extend beyond the payload by 15% of the maximum height and width if the mechanism has any parts that are separate from the payload. For example, if there is a rope that extends beyond the satellite upwards to the drone, this is allowed. As another example, if there is a latch or hook in which the main "body" or portion of the release mechanism remains within the satellite, it is allowed.

Appendix D: Additional Information on PDR

Here are some additional details regarding the PDR, including expectations or clarifications on what is needed to be placed on the PDR:

- **Slide count:** The slide count must be at minimum the amount of slides that are provided in the template. You may add additional slides, to a limit of 5 slides per provided template slide. However, make sure that your slides are able to clearly and concisely convey the design choices your team is making.
- **Battery:** The battery is selected by the team based on research and power calculations in addition to the requirements of the mission guide. Lithium ion batteries are allowed. Do not select counterfeit/non-certified/unverified batteries.
- **Software slides:** Information on software does not need to include pictures of software or completely developed code. Flowcharts and diagrams to represent system processes or how data is being transferred is helpful in conveying to the judges how the software will function. For any GUI-related slides, a simple graphic design layout is fine to convey how you will be displaying your information. You may also use screenshots or pictures from the internet to represent how you will receive your data, for example showing a picture of a console on the platform you intend to use.

Appendix E: Datasheets and Resources for Parts

Below are some of the specifications for the electronic components/sensors as well as some other items provided/recommended by the STECI team:

Equipment	Purchase Link Used	Datasheet/Links/Resources
PETG-HF Filament	Link	• https://bambulab.com/en-ca/filament/petg-hf • https://wiki.bambulab.com/en/filament/petg
ABS Filament	Link	• https://bambulab.com/en-ca/filament/abs • https://wiki.bambulab.com/en/filament/abs_asa_pc
NRF24L01 Set	Link	• https://www.handsontec.com/dataspecs/module/NRF24L01+PA.pdf • https://github.com/tarantula3/NRF24L01-Tutorial- • https://lastminuteengineers.com/nrf24l01-arduino-wireless-communication/
MPU-6050	Link	• https://www.handsontec.com/dataspecs/MPU6050.pdf • https://mysii.gorriens.net/images/arduino/capteurs/gy-521_mpu-6050_3-axis_gyroscope_and_acceleration_sensor_en.pdf • https://www.youtube.com/watch?v=c_BmS2f04X4
BMP280	Link	• https://www.csdcms.ca/Tutorials/5_BMP280b.pdf • https://handsontec.com/dataspecs/sensor/BMP280.pdf • https://components101.com/sensors/gy-bmp280-module • https://lastminuteengineers.com/bme280-arduino-tutorial/
Parachute	N/A	• Cross Parachute making tutorial: https://www.nakka-rocketry.net/xchute

		1.html • https://fruitychutes.com/help_for_parachutes/drone-parachute-tutorials//how_to_make_a_parachute • https://www.sunward1.com/imagespara/The%20Mathematics%20of%20Parachutes(Rev2).pdf
Micro Servos (SG90)	Link	• http://www.ee.ic.ac.uk/pcheung/teaching/DE1_EE/stores/sg90_datasheet.pdf
Battery	N/A (dependent on each team)	• https://startingelectronics.org/beginners/components/batteries/ • https://startingelectronics.org/articles/arduino/battery-powering-arduino-uno/ • https://learnabout-electronics.org/PSU/psu31.php • https://learnabout-electronics.org/PSU/psu32.php